

$A = QR \rightarrow$ Least Squares.

S.V.D (Singular Value Decomposition)

$$A_{m \times n} \quad U^T A V = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_p) = \Sigma \quad p = \min\{m, n\}$$

$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > \sigma_{r+1} = \dots = \sigma_p = 0$

$m > n$

$$A = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$
$$A = U \Sigma V^T$$

$$A = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \Sigma_{11} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_1 & V_2 \end{bmatrix}^T = U_1 \Sigma_{11} V_1^T$$

$$\text{Imag}(A) = \text{span}\{U_1\}$$

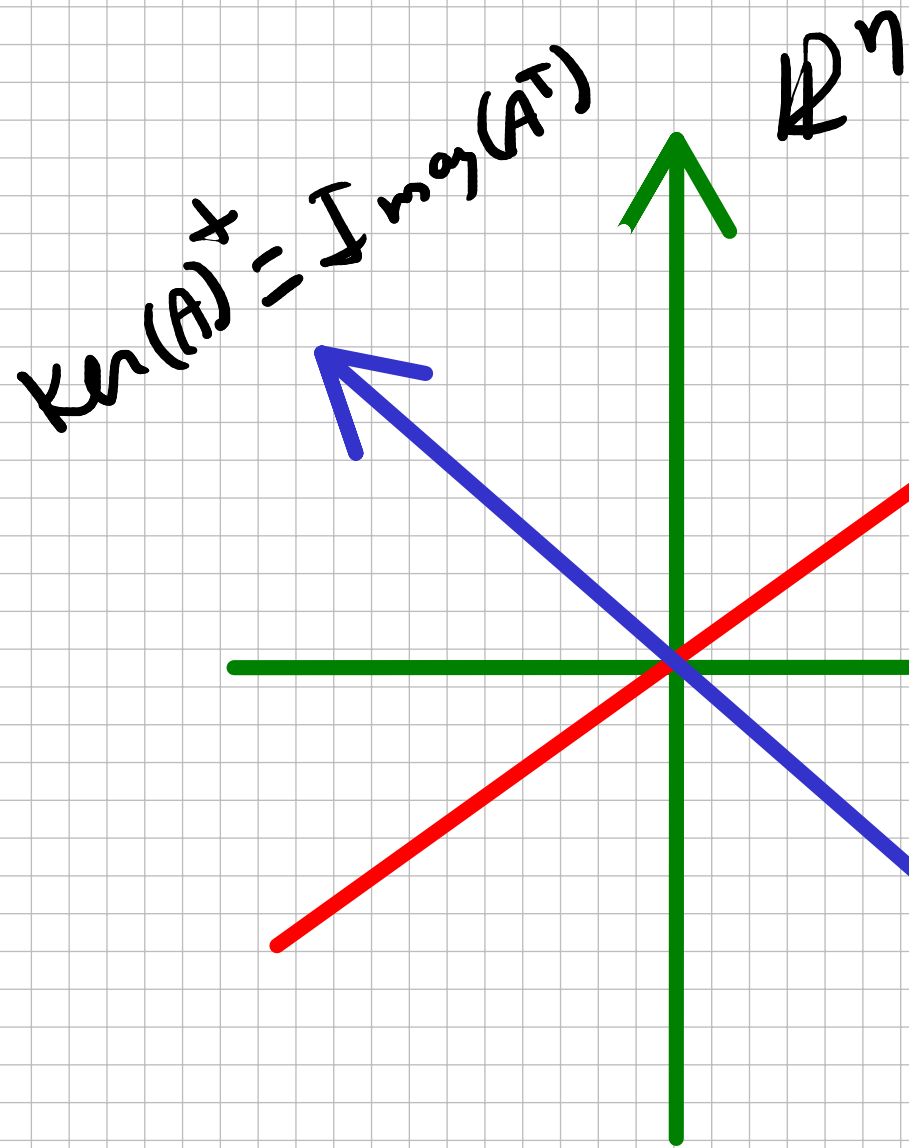
$$\text{Imag}(A)^\perp = \text{span}\{U_2\}$$

$$\ker(A) = \text{span}\{V_2\}$$

$$\ker(A)^\perp = \text{span}\{V_1\}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & 0 \end{pmatrix}_{n \times n}$$

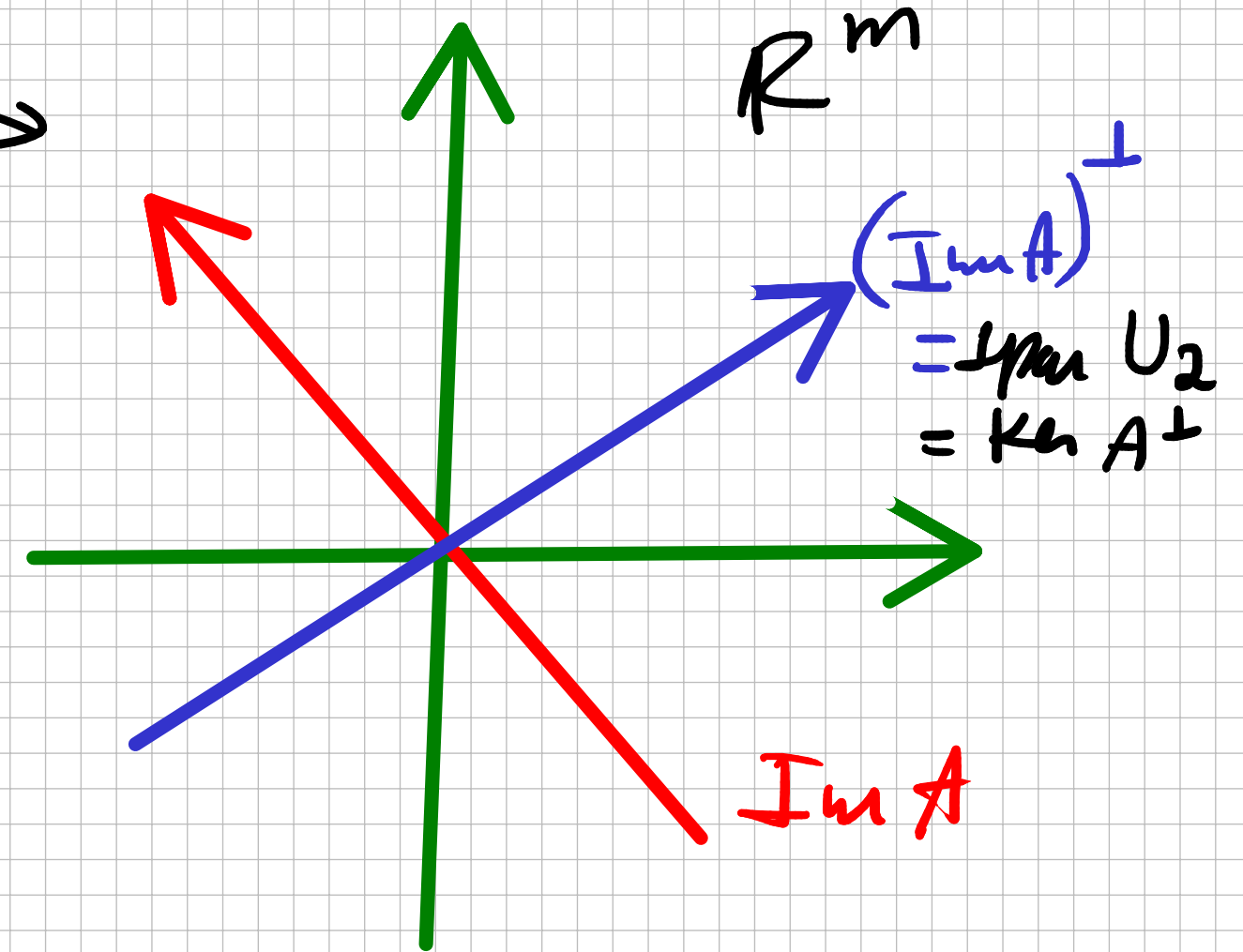
$$A V_2 = U_1 \Sigma_{11} \underbrace{V_1^T V_2}_{=0} = 0$$



A

$\ker A$

A^T



$$A = U \Sigma V^T$$

$$A^T = V \Sigma^T U^T = \underset{\uparrow}{V_1} \Sigma_{11} U_1^T$$

$$A = \sum_{l=1}^{p,r} \underline{\sigma_l} u_l v_l^T$$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$$

$$A = U \Sigma V^T = U_1 \Sigma_{11} V_1^T$$

$$A^+ = V_1 \Sigma_{11}^{-1} U_1^T = \underset{n \times n}{V} \underset{m \times n}{\Sigma^+} \underset{m \times n}{U^T}$$

- Inversa Generalizada
- Pseudo inversa

$$x_{LS} = A^+ b$$

$$U = [u_1 \ u_2 \ \dots \ u_r \ u_{r+1} \ \dots \ u_m]_{m \times m}$$

$$V = [v_1 \ v_2 \ \dots \ v_n]_{n \times n}$$

$$\Sigma^+ = \begin{pmatrix} \overset{r}{\Sigma_{11}^{-1}} & \overset{m-r}{0} \\ \underset{r}{0} & \underset{m-r}{0} \end{pmatrix}_{n \times m}$$

$$\Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r & & \\ & & & & 0 & \dots \end{pmatrix}_{m \times n}$$

$$A^+ = (A^T A)^{-1} A^T \quad m \geq n$$

Para $A = U \Sigma V^T = \sum_{l=1}^r \sigma_l u_l v_l^T$
 $\sigma_1 \geq \sigma_2 \geq \dots$

$$\|B\|_2 = \max_{x \neq 0} \frac{\|Bx\|_2}{\|x\|_2} = \sigma_1(B)$$

$$A_k = \sum_{l=1}^k \sigma_l u_l v_l^T$$

i) $\|A - A_k\|_2 = \left\| \sum_{l=k+1}^r \sigma_l u_l v_l^T \right\|_2 = \sigma_{k+1}$

$$A_k = U \begin{pmatrix} \sigma_1 & & & & 0 \\ & \ddots & & & \\ & & \sigma_k & & 0 \\ & & & \ddots & \\ 0 & & & & 0 \end{pmatrix} V^T$$

$$A - A_k = U \begin{pmatrix} 0 & & & & 0 \\ & \ddots & & & \\ & & \sigma_{k+1} & & \\ & & & \ddots & \\ 0 & & & & 0 \end{pmatrix} V^T$$

$$\text{rango}(A_k) = \#\{\text{valores singulares de } A_k \neq 0\} = k.$$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \dots \geq \sigma_k >> \sigma_{k+1}$$

JP6

$A_{n \times m} = n^2$

$$A_k = \underbrace{[u_1 \dots u_k]}_{m \times k} \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{pmatrix} \underbrace{[v_1 \dots v_k]^T}_{n \times k}$$

$m \times k + n \times k + k = 2k(n+1)$

$$i) M, \text{rang}(M)=k, \quad \|A-M\|_2 \geq \sigma_{k+1}$$

$$i)+ii) \Rightarrow A_k \in \arg \min_{\text{rang}(M)=k} \|A-M\|_2$$

$$\begin{aligned} & \text{M de rango } k, \\ & \dim \text{Im}(M) = k \\ & \dim \text{Ker } M = \underline{n-k} \end{aligned}$$

$$\begin{aligned} & 0 \neq x \in \text{Ker } M \cap \text{span}\{v_1, v_2, \dots, v_k, v_{k+1}\} \\ & \downarrow \text{as no varia por contro de dimensão} \\ & \|x\|_2 = 1 \\ & x = (x^T v_1) v_1 + (x^T v_2) v_2 + \dots + (x^T v_{k+1}) v_{k+1} \end{aligned}$$

$$A v_i = \sigma_i u_i$$

$$\begin{aligned} & \sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_{k+1} \\ & (x^T v_1)^2 + (x^T v_2)^2 + \dots + (x^T v_{k+1})^2 = 1 \end{aligned}$$

$$\|x\|_2 = 1$$

$$\rightarrow y = (A-M)x = Ax - \cancel{Mx}^0$$

$$= (x^T v_1) \sigma_1 u_1 + (x^T v_2) \sigma_2 u_2 + \dots + (x^T v_{k+1}) \sigma_{k+1} u_{k+1}$$

$$\|y\|_2^2 = (x^T v_1)^2 \sigma_1^2 + (x^T v_2)^2 \sigma_2^2 + \dots + \underline{(x^T v_{k+1})^2 \sigma_{k+1}^2} \geq \sigma_{k+1}^2$$

$$\|A - M\|_2 = \max_{z \neq 0} \frac{\|(A - M)z\|_2}{\|z\|_2} \geq \frac{\|(A - z)x\|_2}{\|x\|_2} = \|y\|_2 \geq \underline{\sigma_{k+1}}.$$

$$\checkmark \left\{ \begin{array}{l} \|A - A_k\|_2 = \sigma_{k+1} \\ \|A - M\|_2 \geq \sigma_{k+1} \end{array} \right. \text{ para } M \text{ con } \text{rango}(M) = k.$$

$$\sigma_k(A+E) \leq \sigma_k(A) + \|E\|_2$$

$$\sigma_k(A+E) \leq \|A+E-M\|_2$$

$$\leq \|A-A_k\|_2 + \|E\|_2$$

$$\leq \sigma_k(A) + \|E\|_2$$

M is a diagonal matrix
of rank k

$$M = A_k A_k^T$$

$$\sigma_k(A) \leq \sigma_k(A+E) \leq \sigma_k(A) + \|E\|_2$$

Valores Singulares $\longleftrightarrow$ Valores Propios.